

10.1 Logic expressions answers

Getting started

Complete the truth tables for the following gates:

1 AND

X	Y	X AND Y
0	0	0
0	1	0
1	0	0
1	1	1

2 OR

X	Y	X OR Y
0	0	0
0	1	1
1	0	1
1	1	1

3 NOT

X	NOT X
0	1
1	0

4 NAND

X	Y	X NAND Y
0	0	1
0	1	1
1	0	1
1	1	0

5 NOR

X	Y	X NOR Y
0	0	1
0	1	0
1	0	0
1	1	0

6 XOR

X	Y	X XOR Y
0	0	0
0	1	1
1	0	1
1	1	0

Practice

1 Evaluate each statement to identify if the output is 1 (true) or 0 (false).

a 1 AND 0 **0**

b 1 OR 1 **1**

c 1 NOR 0 **0**

d 1 XOR 1 **0**

e NOT(1) *0*

f 1 NAND 0 *1*

g 1 AND 1 *1*

h 0 OR 0 *0*

i 0 NOR 0 *1*

j 1 XOR 0 *1*

2 Complete the truth table for the following logic expressions.

a

A	B	C	A AND B AND C
0	0	0	<i>0</i>
0	0	1	<i>0</i>
0	1	0	<i>0</i>
0	1	1	<i>0</i>
1	0	0	<i>0</i>
1	0	1	<i>0</i>
1	1	0	<i>0</i>
1	1	1	<i>1</i>

b

A	B	C	A XOR (B OR C)
0	0	0	<i>0</i>
0	0	1	<i>1</i>
0	1	0	<i>1</i>
0	1	1	<i>1</i>
1	0	0	<i>1</i>
1	0	1	<i>0</i>
1	1	0	<i>0</i>
1	1	1	<i>0</i>

c

A	B	C	(A AND B) OR (C XOR B)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

d

A	B	C	(NOT A NAND B) NOR C
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

e

A	B	C	(C XOR A) AND (B OR A)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Challenge

Write a logic expression that makes use of **four** inputs; A, B, C and D.

Create the truth table for your logic expression.

Swap the expression with another learner and draw the truth table for their logic expression.

Compare your truth tables and see if they match.

Student answers will vary.

10.2 Logic circuits answers

Getting started

Draw each of the logic gate symbols.

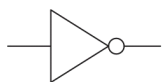
1 AND



2 OR



3 NOT



4 NAND



5 NOR



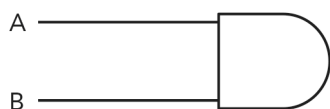
6 XOR



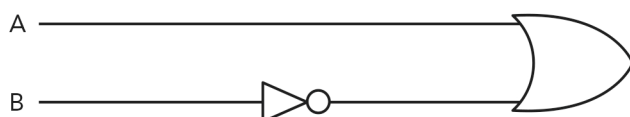
Practice

1 Draw the logic circuit for the following logic statements:

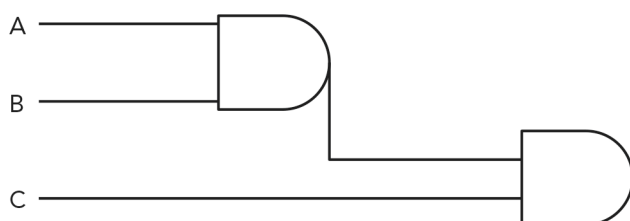
a A AND B



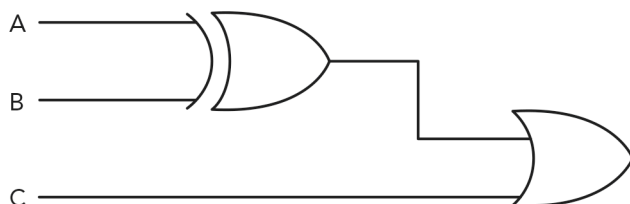
b A OR NOT B



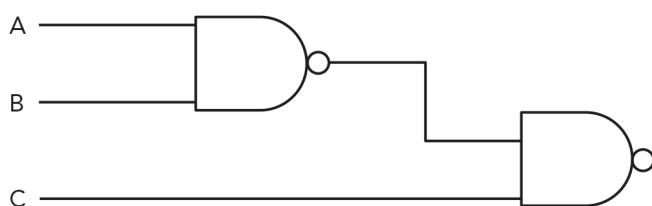
c A AND B AND C



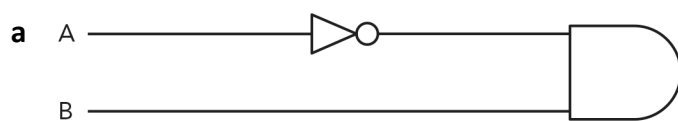
d (A XOR B) OR C



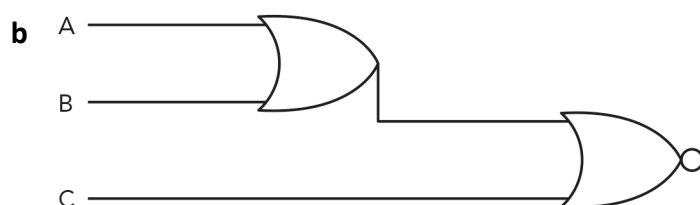
e A NAND B NAND C



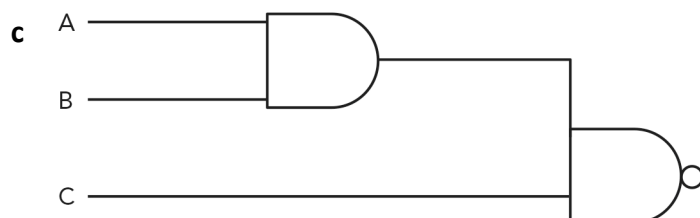
2 Write the logic expression for the following logic circuits



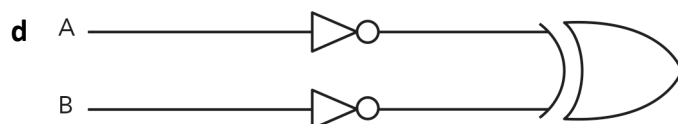
NOT A AND B



(A OR B) NOR C



(A AND B) NAND C



NOT A XOR NOT B

Challenge

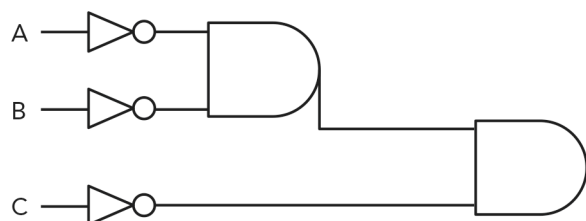
Write the logic statement for each of the truth tables **and** draw its logic circuit.

Answers – these are examples only, there are different expressions and circuits that will output the same result.

1

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

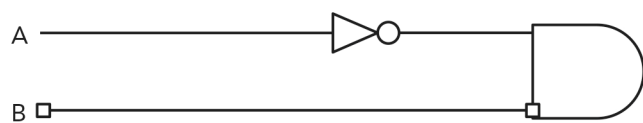
Statement: NOT A AND NOT B AND NOT C



2

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

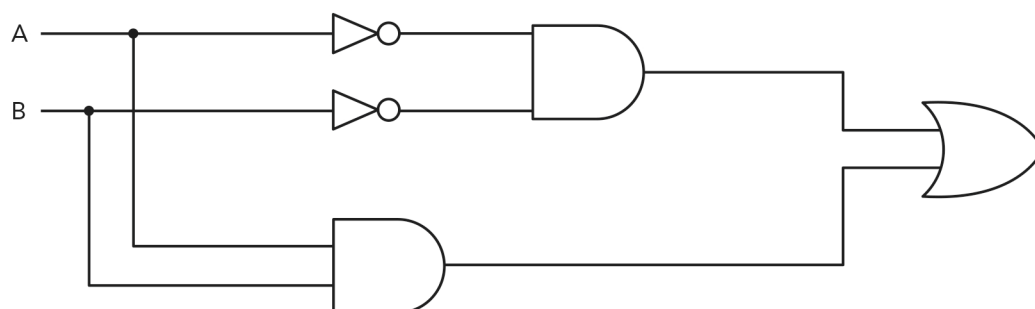
Statement: *NOT A AND B*



3

A	B	C	Output
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

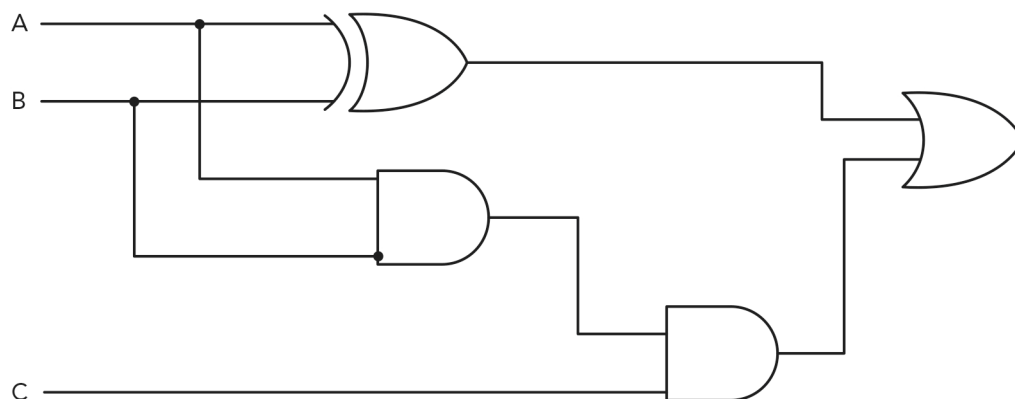
(*NOT A AND NOT B*) OR (*A AND B*)



4

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Statement: $(B \text{ XOR } C) \text{ OR } (A \text{ AND } B \text{ AND } C)$



10.3 Logic answers

Getting started

Complete the following sentences:

The _____ **AND** _____ gate only returns 1 when both inputs are 1.

The _____ **OR** _____ gate only returns 0 when both inputs are 0.

The _____ **NOT** _____ gate only returns 1 if the input is 0.

The _____ **XOR** _____ gate returns 0 if both inputs are 0, and when both inputs are 1.

The _____ **NOR** _____ gate only returns 1 if both inputs are 0.

Practice

Correct the error(s) in the truth tables.

1

A	B	C	A AND (B OR C)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

2

A	B	C	A XOR NOT B
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

3

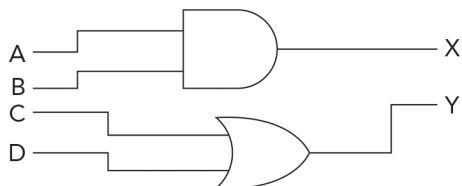
A	B	C	(A NOR B) OR (C AND A)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Challenge

Draw a logic diagram for the following 2-output statements.

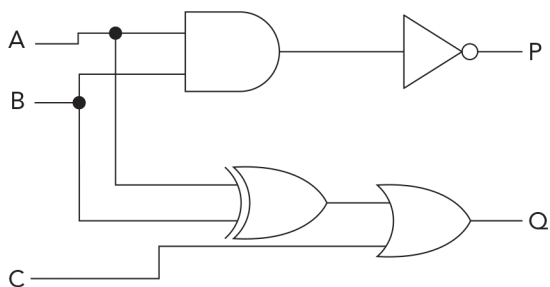
1 $X = A \text{ AND } B$

$Y = C \text{ OR } D$



2 $P = \text{NOT } (A \text{ AND } B)$

$Q = (A \text{ XOR } B) \text{ OR } C$



3 $S = (A \text{ NOR } B) \text{ AND } C$

$T = \text{NOT}(A \text{ OR } B)$

